RUTUJA KADAM

DAY 06 – IPv4 ADDRESSING & CONFIGURATION

IPv4 THEORY (10 Marks)

1. What is IPv4?

IPv4 (Internet Protocol Version 4) is a protocol that provides logical addressing to devices communicating over IP networks. It helps identify devices and enables data communication between them.

2. How many bits are in an IPv4 address?

An IPv4 address contains **32 bits**, divided into four groups of 8 bits each.

3. What is an octet?

An octet is a group of 8 bits in an IPv4 address. Each octet is represented by a decimal number ranging from 0 to 255.

4. How many octets are in an IPv4 address?

An IPv4 address contains **four octets**, separated by dots.

Example: 192.168.1.10

Here, 192, 168, 1 and 10 are the four octets.

5. What is a subnet mask?

A subnet mask is used to identify which portion of an IPv4 address represents the network and which portion represents the host.

Example: 255.255.255.0 (/24).

6. Difference between network portion and host portion?

Network Portion	Host Portion
Identifies the network.	Identifies a device within the network.
It is used to determine which network a device belongs to.	It identifies a particular device on that network.
Example: 192.168.1	Example: 10

Example: For IP address 192.168.1.10 with subnet mask 255.255.255.0, the network portion is 192.168.1 and the host portion is 10.

7. What is a private IPv4 address?

A private IPv4 address is an IP address used within local networks such as homes, offices and schools. It is not directly routable on the public Internet.

8. Write all three private IPv4 ranges.

The three private IPv4 address ranges are:

1. **10.0.0.0/8:** 10.0.0.0 – 10.255.255.255
2. **172.16.0.0/12:** 172.16.0.0 – 172.31.255.255
3. **192.168.0.0/16:** 192.168.0.0 – 192.168.255.255

9. What is a public IPv4 address?

A public IPv4 address is a globally routable IP address used for communication over the Internet. Internet-facing systems use public IP addresses to communicate with other networks.

10. What is the IPv4 loopback address?

The IPv4 loopback address is 127.0.0.1, also known as localhost. It allows a computer to communicate with itself and test its local TCP/IP network stack.

11. What is APIPA?

APIPA (Automatic Private IP Addressing) is a feature that automatically assigns an IP address to a device when it cannot obtain an IP address from a DHCP server.

Its address range is 169.254.0.0/16.

12. What does 169.254.x.x commonly indicate?

A 169.254.x.x address commonly indicates that the computer could not obtain an IP address from the DHCP server and has automatically assigned itself an APIPA address.

13. What is a default gateway?

A default gateway is usually a router that forwards network traffic from a local network to other networks, including the Internet.

Example: 192.168.1.1.

14. Static vs dynamic IP addressing?

Static IP Addressing	**Dynamic IP Addressing**
IP address is configured manually.	IP address is assigned automatically.
It remains consistent until changed.	It may change over time.

Commonly used for servers, printers and network equipment.	Commonly used for computers and mobile devices.
Does not require automatic DHCP assignment.	Usually uses DHCP for assignment.

15. What is DHCP?

DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP configuration to devices. It provides an IP address, subnet mask, default gateway and DNS server.

16. What is a network address?

A network address identifies the entire network rather than an individual device.

Example: For 192.168.1.10/24, the network address is 192.168.1.0.

17. What is a broadcast address?

A broadcast address is used to send data to all devices within a particular network or broadcast domain.

Example: For the network 192.168.1.0/24, the broadcast address is 192.168.1.255.

18. Why is IPv4 knowledge important in cybersecurity?

IPv4 knowledge is important in cybersecurity because:

1. It helps identify source and destination IP addresses.
2. It helps analyse firewall logs and server logs.
3. It helps investigate network traffic and packet captures.
4. It supports IDS/IPS alert analysis.
5. It helps understand Nmap scan results.
6. It helps analyse Wireshark network traffic.
7. It helps identify suspicious network communication.
8. It supports network monitoring and incident response.

Practical: Identify Your Own Network Configuration

1. **IPv4 Address:** 192.168.0.101
2. **Subnet Mask:** 255.255.255.0
3. **Default Gateway:** 192.168.0.1

4. **DNS Server:** 192.168.0.1
5. **DHCP Status:** Enabled (Yes)

- **Network Adapter Name:** Wireless LAN adapter Wi-Fi
- **Adapter Description:** MediaTek Wi-Fi 6E MT7922 (RZ616) 160MHz PCIe Adapter

➢ **SCREENSHOTS**

1)ipconfig/all



2)ipconfig

```
C:\Users\HP>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::49b1:753c:2f33:8f5e%8
   IPv4 Address. . . . . . . . . . . : 192.168.0.101
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\HP>
```

3)ping

```
C:\Users\HP> ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\HP>
```